Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits
INT108	PYTHON PROGRAMMING	3	0	2	4
Course Weightage	ATT: 5 CA: 50 ETP: 45				
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT				

Course Outcomes :Through this course students should be able to

CO1 :: describe the installation of python environment and basics of Python language

CO2 :: apply the conditional and iterative statements for evaluating the appropriate alternates

CO3 :: explore functions, including recursion, with parameters and arguments in Python.

CO4 :: construct the core data structures like lists, dictionaries, tuples and sets in Python to store, process and sort the data

CO5 :: apply the concepts of Object-oriented programming as used in Python using encapsulation, polymorphism, and inheritance

CO6 :: examine file handling operations and effectively apply regular expressions for pattern matching

	TextBooks (T)		
Sr No	**Title**	**Author**	**Publisher Name**
T-1	FUNDAMENTALS OF PYTHON –FIRST PROGRAM	KENNETH A. LAMBERT	CENGAGE LEARNING

	Reference Books (R)		
Sr No	**Title**	**Author**	**Publisher Name**
R-1	PYTHON PROGRAMMING: USING PROBLEM SOLVING APPROACH	REEMA THAREJA	OXFORD UNIVERSITY PRESS

Relevant Websites (RW)		
Sr No	**(Web address) (only if relevant to the course)**	**Salient Features**
RW-1	https://www.datacamp.com/courses/intro-to-python-for-data-science	Python course
RW-2	https://www.w3schools.com/python/python_tuples.asp	Python Tuples
RW-3	https://www.coursera.org/learn/python Learn	Python from basics

Audio Visual Aids (AV)		
Sr No	**(AV aids) (only if relevant to the course)**	**Salient Features**
AV-1	https://nptel.ac.in/courses/106106145	Python course

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Software/Equipments/Databases		
Sr No	**(S/E/D) (only if relevant to the course)**	**Salient Features**
SW-1	https://www.python.org/download	Python software
SW-2	https://anaconda.org/anaconda/python	Anaconda software Installation

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Setting up your Programming Environment (Python versions)	T-1	SW-1 SW-2	Lecture 1 should be used to discuss Lecture zero.	Student will understand the use of python programming and its importance in industry.	Class room discussion using power point presentation and blended learning using CodeTantra	
	Lecture 2	Setting up your Programming Environment (Python on windows)	T-1 R-1	RW-1 RW-3 AV-1	Lecture 2 to discuss the python program concepts	Student will understand the use of python programming and its importance in industry.	Class room discussion using power point presentation and Blended Learning using CodeTantra	
		Setting up your Programming Environment (running a ‘Hello World’ program)	T-1 R-1	RW-1 RW-3 AV-1	Lecture 2 to discuss the python program concepts	Student will understand the use of python programming and its importance in industry.	Class room discussion using power point presentation and blended learning using CodeTantra	

Week 1	Lecture 3	Variables, Expression and Statements:(Naming and using variables)		RW-1 RW-3 AV-1	Lecture 3 to discuss about variables, values and types.	Student will learn how to declare a variable, rules to follow for naming a variable and how to assign values to variables	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Variables, Expression and Statements:(Avoiding Name Error when using variables)		RW-1 RW-3 AV-1	Lecture 3 to discuss about variables, values and types.	Student will learn how to declare a variable, rules to follow for naming a variable and how to assign values to variables	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Variables, Expression and Statements:(Values and types)		RW-1 RW-3 AV-1	Lecture 3 to discuss about variables, values and types.	Student will learn how to declare a variable, rules to follow for naming a variable and how to assign values to variables	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Variables, Expression and Statements:(variables)		RW-1 RW-3 AV-1	Lecture 3 to discuss about variables, values and types.	Student will learn how to declare a variable, rules to follow for naming a variable and how to assign values to variables	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 2	Lecture 4	Variables, Expression and Statements:(variables name and keywords)		RW-1 RW-3 AV-1	Lecture 5 and 6 to discuss about keywords, operators and precedence of operators and statements	Student will learn about keywords, different arithmetic operators and their precedence and statements	Class room discussion using power point presentation and blended learning using CodeTantra	
		Variables, Expression and Statements:(statements)		RW-1 RW-3 AV-1	Lecture 5 and 6 to discuss about keywords, operators and precedence of operators and statements	Student will learn about keywords, different arithmetic operators and their precedence and statements	Class room discussion using power point presentation and brain stroming by solving problems on CodeTantra	
		Variables, Expression and Statements:(operators and operand)		RW-1 RW-3 AV-1	Lecture 5 and 6 to discuss about keywords, operators and precedence of operators and statements	Student will learn about keywords, different arithmetic operators and their precedence and statements	Class room discussion using power point presentation and Live demonstration of programs in Python	

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Week 2	Lecture 4	Variables, Expression and Statements:(order of operations)		RW-1 RW-3 AV-1	Lecture 5 and 6 to discuss about keywords, operators and precedence of operators and statements	Student will learn about keywords, different arithmetic operators and their precedence and statements	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 5	Variables, Expression and Statements:(variables name and keywords)		RW-1 RW-3 AV-1	Lecture 5 and 6 to discuss about keywords, operators and precedence of operators and statements	Student will learn about keywords, different arithmetic operators and their precedence and statements	Class room discussion using power point presentation and blended learning using CodeTantra	
		Variables, Expression and Statements:(statements)		RW-1 RW-3 AV-1	Lecture 5 and 6 to discuss about keywords, operators and precedence of operators and statements	Student will learn about keywords, different arithmetic operators and their precedence and statements	Class room discussion using power point presentation and brain stroming by solving problems on CodeTantra	
		Variables, Expression and Statements:(operators and operand)		RW-1 RW-3 AV-1	Lecture 5 and 6 to discuss about keywords, operators and precedence of operators and statements	Student will learn about keywords, different arithmetic operators and their precedence and statements	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Variables, Expression and Statements:(order of operations)		RW-1 RW-3 AV-1	Lecture 5 and 6 to discuss about keywords, operators and precedence of operators and statements	Student will learn about keywords, different arithmetic operators and their precedence and statements	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 6	Variables, Expression and Statements:(operations on string)		RW-1 AV-1	Lecture 6 to discuss about operations on strings, comments and composition	Students will learn about strings and operations on string and how to give single and multi line comment in a program.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Variables, Expression and Statements:(composition and comments)		RW-1 AV-1	Lecture 6 to discuss about operations on strings, comments and composition	Students will learn about strings and operations on string and how to give single and multi line comment in a program.	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 3	Lecture 7	Conditional statements (modulus operator)		RW-1 RW-3 AV-1	Lecture 7 and 8 to discuss modulus operator, conditional operators and logic operators and random numbers	Students will learn about modulus operator, conditional operators and logical operators and random numbers.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Conditional statements (Random numbers)		RW-1 RW-3 AV-1	Lecture 7 and 8 to discuss modulus operator, conditional operators and logic operators and random numbers	Students will learn about modulus operator, conditional operators and logical operators and random numbers.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Conditional statements (Boolean expressions)		RW-1 RW-3 AV-1	Lecture 7 and 8 to discuss modulus operator, conditional operators and logic operators and random numbers	Students will learn about modulus operator, conditional operators and logical operators and random numbers.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Conditional statements(logic operators)		RW-1 RW-3 AV-1	Lecture 7 and 8 to discuss modulus operator, conditional operators and logic operators and random numbers	Students will learn about modulus operator, conditional operators and logical operators and random numbers.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 8	Conditional statements (modulus operator)		RW-1 RW-3 AV-1	Lecture 7 and 8 to discuss modulus operator, conditional operators and logic operators and random numbers	Students will learn about modulus operator, conditional operators and logical operators and random numbers.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Conditional statements (Random numbers)		RW-1 RW-3 AV-1	Lecture 7 and 8 to discuss modulus operator, conditional operators and logic operators and random numbers	Students will learn about modulus operator, conditional operators and logical operators and random numbers.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Conditional statements (Boolean expressions)		RW-1 RW-3 AV-1	Lecture 7 and 8 to discuss modulus operator, conditional operators and logic operators and random numbers	Students will learn about modulus operator, conditional operators and logical operators and random numbers.	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 3	Lecture 8	Conditional statements(logic operators)		RW-1 RW-3 AV-1	Lecture 7 and 8 to discuss modulus operator, conditional operators and logic operators and random numbers	Students will learn about modulus operator, conditional operators and logical operators and random numbers.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 9	Conditional statements (conditional)		RW-1 RW-3 AV-1	Lecture 9 and 10 to discuss conditional statements and nested conditional statements	Students will learn about if else, if elif else statements and nested if else statements.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Conditional statements (nested conditionals)		RW-1 RW-3 AV-1	Lecture 9 and 10 to discuss conditional statements and nested conditional statements	Students will learn about if else, if elif else statements and nested if else statements.	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 4	Lecture 10	Conditional statements (conditional)		RW-1 RW-3 AV-1	Lecture 9 and 10 to discuss conditional statements and nested conditional statements	Students will learn about if else, if elif else statements and nested if else statements.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Conditional statements (nested conditionals)		RW-1 RW-3 AV-1	Lecture 9 and 10 to discuss conditional statements and nested conditional statements	Students will learn about if else, if elif else statements and nested if else statements.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 11	Iterative statements(while statements)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Iterative statements(for loop statement)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 4	Lecture 11	Iterative statements(Nested for)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Iterative statements(Nested while)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Iterative statements(Random numbers in loops)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Iterative statements (encapsulation and generalization)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 12	Iterative statements(while statements)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Iterative statements(for loop statement)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Iterative statements(Nested for)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 4	Lecture 12	Iterative statements(Nested while)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Iterative statements(Random numbers in loops)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Iterative statements (encapsulation and generalization)		RW-1 RW-3 AV-1	Lecture 11 and 12 to discuss about for loop and while loop, nested looping statements and random numbers in loop	Student will learn to declare for and while loop, nested looping statements and encapsulation and generalization	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 5	Lecture 13	Functions and recursion (math functions)		RW-1 RW-3 AV-1	Lecture 13 to discuss about functions, its importance and some predefined math function.	Students will learn about use of functions in programs and different math function in python	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 14	Functions and recursion(type conversion and coercion)		RW-1 RW-3 AV-1	Lecture 14 to discuss about type coercion and type conversion.	Students will learn how data is converted internally during certain calculations and sometimes we need to enforce type conversion in order to obtain results.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 15	Functions and recursion (adding new function)		RW-1 RW-3 AV-1	Lecture 15 to discuss about user defined functions.	Students will learn how to create functions in python.	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 6	Lecture 16				Test - Code based 1			

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Week 6	Lecture 17	Functions and recursion (function calls)		RW-1 RW-3 AV-1	Lecture 17 to discuss about function calls.	Students will learn about how to call a function once defined, return statements in function.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 18	Functions and recursion (parameters and argument)		RW-1 RW-3 AV-1	Lecture 18 to discuss about arguments and parameters	Students will learn how to declare function with arguments, default arguments and non default arguments, order of default and non default arguments	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 7	Lecture 19	Functions and recursion (recursion and its use)		RW-1 RW-3 AV-1	Lecture 19 to discuss about recursion and its use.	Student will learn about concept of recursion and its use by implementing factorial and Fibonacci programs	Class room discussion using power point presentation and Live demonstration of programs in Python	
		SPILL OVER						
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
		MID-TERM						
Week 8	Lecture 22	String(string a compound data type)		RW-1 RW-3 AV-1	Lecture 22 to discuss about string declaration, length of string, string traversal and string slicing.	Students will learn how to declare a string, len()function, postive and negative indexing and slicing operator.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		String(length)		RW-1 RW-3 AV-1	Lecture 22 to discuss about string declaration, length of string, string traversal and string slicing.	Students will learn how to declare a string, len()function, postive and negative indexing and slicing operator.	Class room discussion using power point presentation and Live demonstration of programs in Python	

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Week 8	Lecture 22	String(string traversal)		RW-1 RW-3 AV-1	Lecture 22 to discuss about string declaration, length of string, string traversal and string slicing.	Students will learn how to declare a string, len()function, postive and negative indexing and slicing operator.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		String(string slices)		RW-1 RW-3 AV-1	Lecture 22 to discuss about string declaration, length of string, string traversal and string slicing.	Students will learn how to declare a string, len()function, postive and negative indexing and slicing operator.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 23	String(comparision)	T-1	RW-1 RW-3	Lecture 23 to discuss comparison, find function , looping and counting	Students will learn how to compare strings, find function, for loop statement using string	Class room discussion using power point presentation and Live demonstration of programs in Python	
		String(find function)	T-1	RW-1 RW-3	Lecture 23 to discuss comparison, find function , looping and counting	Students will learn how to compare strings, find function, for loop statement using string	Class room discussion using power point presentation and Live demonstration of programs in Python	
		String(looping and counting)	T-1	RW-1 RW-3	Lecture 23 to discuss comparison, find function , looping and counting	Students will learn how to compare strings, find function, for loop statement using string	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 24	Lists(list values)	R-1	RW-1 AV-1	Lecture 24 to discuss about lists, membership operator, length function	Students will learn about list data structure, how to declare list, len() function and in operator.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Lists(length)	R-1	RW-1 AV-1	Lecture 24 to discuss about lists, membership operator, length function	Students will learn about list data structure, how to declare list, len() function and in operator.	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 8	Lecture 24	Lists(membership)	R-1	RW-1 AV-1	Lecture 24 to discuss about lists, membership operator, length function	Students will learn about list data structure, how to declare list, len() function and in operator.	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 9	Lecture 25	Lists(operations)	T-1	RW-1 AV-1	Lecture 25 to discuss about indexing and slicing of list, nested list and different operation on list	Students will learn to create nested list, indexing of list, slicing of list, for loop using list and different operations on List like insertion, deletion and substitution	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Lists(slices, deletion)	T-1	RW-1 AV-1	Lecture 25 to discuss about indexing and slicing of list, nested list and different operation on list	Students will learn to create nested list, indexing of list, slicing of list, for loop using list and different operations on List like insertion, deletion and substitution	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Lists(accessing elements, list and for loops, list parameters and nested list)	T-1	RW-1 AV-1	Lecture 25 to discuss about indexing and slicing of list, nested list and different operation on list	Students will learn to create nested list, indexing of list, slicing of list, for loop using list and different operations on List like insertion, deletion and substitution	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 26	Tuples and Dictionaries (mutability and tuples)		RW-2 RW-3	Lecture 26 to discuss about tuples, mutable vs immutable, tuple assignment and tuple as return value.	Students will learn about tuples, difference between mutable and immutable data type, tuple as return type	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Tuples and Dictionaries (tuple assignment, tuple as return values)		RW-2 RW-3	Lecture 26 to discuss about tuples, mutable vs immutable, tuple assignment and tuple as return value.	Students will learn about tuples, difference between mutable and immutable data type, tuple as return type	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 9	Lecture 27	Tuples and Dictionaries (dictionaries operations and methods, sparse matrices, aliasing and coping)		RW-3	Lecture 27 to discuss about dictionaries operations and methods, sparse matrices, aliasing and coping	Students will learn about dictionaries data type, accessing elements of dictionary, operations on dictionaries, aliasing and copying	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 10	Lecture 28	Classes and objects(Creating classes)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Classes and objects(creating instance objects)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Classes and objects (accessing attributes)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 29				Test - Code based 2			
		Classes and objects(Creating classes)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 10	Lecture 29	Classes and objects(creating instance objects)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Classes and objects (accessing attributes)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 30	Classes and objects(Creating classes)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Classes and objects(creating instance objects)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Classes and objects (accessing attributes)	T-1	RW-1 AV-1	Lecture 28 and 30 to discuss about classes, objects and accessing attributes	Students will learn how to declare a class in python, constructor of a class, member functions, instance vs class variables in a class, creating objects and accessing attributes in class	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 11	Lecture 31	Object oriented programming terminology (Class Inheritance)	T-1	RW-1 AV-1	Lecture 31 to discuss about concept of inheritance	Student will learn about inheritance, different types of inheritance in python.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 32	Object oriented programming terminology (Overriding Methods)		RW-1 AV-1	Lecture 32 to discuss about overriding methods and data hiding.	Students will learn the importance of data hiding, private members of class, accessing private members outside the class.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Object oriented programming terminology (Data Hiding)		RW-1 AV-1	Lecture 32 to discuss about overriding methods and data hiding.	Students will learn the importance of data hiding, private members of class, accessing private members outside the class.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 33	Object oriented programming terminology (Function Overloading)		RW-1 AV-1	Lecture 33 to discuss the concept of function overloading and operator overloading	Students will learn about function overloading and operator overloading	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 12	Lecture 34	Files and Exceptions(text files, writing variables)		RW-1 AV-1	Lecture 34 and 35 to discuss about text files, Reading from a file, writing to a file	Students will learn the concept of file handling how to read a text file and how to write in text file.	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Files and Exceptions (Reading from a file, writing to a file)		RW-1 AV-1	Lecture 34 and 35 to discuss about text files, Reading from a file, writing to a file	Students will learn the concept of file handling how to read a text file and how to write in text file.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 35	Files and Exceptions(text files, writing variables)		RW-1 AV-1	Lecture 34 and 35 to discuss about text files, Reading from a file, writing to a file	Students will learn the concept of file handling how to read a text file and how to write in text file.	Class room discussion using power point presentation and Live demonstration of programs in Python	

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Week 12	Lecture 35	Files and Exceptions (Reading from a file, writing to a file)		RW-1 AV-1	Lecture 34 and 35 to discuss about text files, Reading from a file, writing to a file	Students will learn the concept of file handling how to read a text file and how to write in text file.	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 36	Files and Exceptions (directories, pickling, handling the zero Division error exception)	R-1	RW-1 RW-3 AV-1	Lecture 36 to discuss about concept of pickling , exception handling in python	Students will learn how to store data without changing the original form using pickling concept, handling exception using try except block	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Files and Exceptions(using try-except blocks, The else block, Handling the File Not found error exception)	R-1	RW-1 RW-3 AV-1	Lecture 36 to discuss about concept of pickling , exception handling in python	Students will learn how to store data without changing the original form using pickling concept, handling exception using try except block	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 13	Lecture 37				Test - Code based 3			
	Lecture 38	Regular Expressions (Concept of regular expression)		RW-1 RW-3 AV-1	Lecture 38 and 39 to discuss about regular expressions, types of regular expression, Web Scraping by using Regular Expressions	Students will learn about Web Scraping by using Regular Expressions	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Regular Expressions(various types of regular expressions)		RW-1 RW-3 AV-1	Lecture 38 and 39 to discuss about regular expressions, types of regular expression, Web Scraping by using Regular Expressions	Students will learn about Web Scraping by using Regular Expressions	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Regular Expressions(using match function)		RW-1 RW-3 AV-1	Lecture 38 and 39 to discuss about regular expressions, types of regular expression, Web Scraping by using Regular Expressions	Students will learn about Web Scraping by using Regular Expressions	Class room discussion using power point presentation and Live demonstration of programs in Python	

Week 13	Lecture 38	Regular Expressions(Web Scraping by using Regular Expressions)		RW-1 RW-3 AV-1	Lecture 38 and 39 to discuss about regular expressions, types of regular expression, Web Scraping by using Regular Expressions	Students will learn about Web Scraping by using Regular Expressions	Class room discussion using power point presentation and Live demonstration of programs in Python	
	Lecture 39	Regular Expressions (Concept of regular expression)		RW-1 RW-3 AV-1	Lecture 38 and 39 to discuss about regular expressions, types of regular expression, Web Scraping by using Regular Expressions	Students will learn about Web Scraping by using Regular Expressions	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Regular Expressions(various types of regular expressions)		RW-1 RW-3 AV-1	Lecture 38 and 39 to discuss about regular expressions, types of regular expression, Web Scraping by using Regular Expressions	Students will learn about Web Scraping by using Regular Expressions	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Regular Expressions(using match function)		RW-1 RW-3 AV-1	Lecture 38 and 39 to discuss about regular expressions, types of regular expression, Web Scraping by using Regular Expressions	Students will learn about Web Scraping by using Regular Expressions	Class room discussion using power point presentation and Live demonstration of programs in Python	
		Regular Expressions(Web Scraping by using Regular Expressions)		RW-1 RW-3 AV-1	Lecture 38 and 39 to discuss about regular expressions, types of regular expression, Web Scraping by using Regular Expressions	Students will learn about Web Scraping by using Regular Expressions	Class room discussion using power point presentation and Live demonstration of programs in Python	
Week 14	Lecture 40				Programming Practice			
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

CA Category of this Course Code is:C020102 (Total 4 tasks, 2 compulsory and out of remaining 1 best out of 2 to be considered)

Component	Iscompulsory	Weightage (%)	Mapped CO(s)
Test - Code based 1	NO	30	CO1, CO2, CO3
Test - Code based 2	NO	30	CO2, CO3, CO4
Test - Code based 3	NO	30	CO1, CO2, CO3, CO4, CO5
Programming Practice	Yes	40	CO1, CO2, CO3, CO4, CO5, CO6

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Programming Practice	To enhance the student problem solving abilities by solving problems on platform from learned concepts	Each student must solve a set of problems(MCQ +Coding problems) as mentioned in the course during the class. These problems are mandatory but non evaluative. For each unit (6 units in total), students must solve 40 randomized problems in each unit in the form of test totaling 240 problems across all units. These problems are weighted which will be considered during final calculation.	Individual	Online	40	2 / 14
Test - Code based 1	To check the students knowledge on the learned topics.	Test will be conducted on CodeTantra Secure application. The test pattern includes 10 multiple-choice questions, each worth 1 mark, and 2 coding questions, each worth 10 marks.	Individual	Online	30	5 / 6
Test - Code based 2	To check the students knowledge on the learned topics.	Test will be conducted on CodeTantra Secure application. The test pattern includes 10 multiple-choice questions, each worth 1 mark, and 2 coding questions, each worth 10 marks.	Individual	Online	30	9 / 10
Test - Code based 3	To check the students knowledge on the learned topics.	Test will be conducted on CodeTantra Secure application. The test pattern includes 10 multiple-choice questions, each worth 1 mark, and 2 coding questions, each worth 10 marks.	Individual	Online	30	12 / 13

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	List of Practical's	1. Program to enter two numbers and print the arithmetic operations like +,-,*, /, // and %.		Student will learn to write a basic program of arithmetic operations in python.
Practical 2	List of Practical's	2. Write a program to find whether an inputted number is perfect or not.		Student will implement a program to check whether a given number is perfect or not.
Practical 3	List of Practical's	3.Write a Program to check if the entered number is Armstrong or not.		Student will implement a Program to check if the entered number is Armstrong or not.
Practical 4	List of Practical's	4.Write a Program to find factorial of the entered number.		Student will implement a program to find factorial of the entered number using for loop.
Practical 5	List of Practical's	5.Write a Program to enter the number of terms and to print the Fibonacci Series.		Student will implement a program to print Fibonacci series.
Practical 6	List of Practical's	6.Write a Program to enter the string and to check if it’s palindrome or not using loop.		Student will learn the logic of palindrome and implement same.
Practical 7	List of Practical's	7.Recursively find the factorial of a natural number.		Student will implement a program for factorial using recurssion.
Practical 8	List of Practical's	10.Read a text file and display the number of vowels/consonants/uppercase/lowercase characters in the file.		Student will implement a program in file handling.
Practical 9	List of Practical's	11.Create a binary file with name and roll no. Search for a given roll number and display the name, if not found display appropriate message.		Student will implement the concept of pickling
Practical 10	List of Practical's	8.Read a file line by line and print it.		Student will understand the concept of file handling and basic operations in file handling.
	List of Practical's	9.Remove all the lines that contain the character “a” in a file and write it into another file.		Student will understand the concept of file handling and basic operations in file handling.
Practical 11	List of Practical's	12.Write a random number generator that generates random numbers between 1 and 6(simulates a dice)		Student will implement a program to generate random numbers.
Practical 12	List of Practical's	13.Write a python program to implement a stack using a list data structure.		Student will implement the structure stack using the list data structure
Practical 13	List of Practical's	14.Take a sample of ten phishing e-mails (any text file) and find most common		Student will implement a program to find the most common fishing email
Practical 14	List of Practical's	15.Read a text file line by line and display each word separated by a #		Student will use the concept of file handling to solve the problem

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	SPILL OVER			
Practical 15	Spill Over			

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